

PhD Math Camp

Ignacio Urbina

Stony Brook University, Fall 2026

Program: PhD in Political Science, Fall 2026

Room: N703.

Dates: *Monday, August 17th – Friday, August 21st*

Meeting blocks: Daily blocks vary; see the schedule in Section III. The camp is scheduled around the mandatory departmental and Graduate School events of that week.

Contact Information

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Office Hours: After class or by appointment.

I Overview

Course Description

This intensive one-week math camp is designed to equip first-year PhD students in political science at Stony Brook University with essential mathematical knowledge needed for advanced methods courses. The aim is to build a foundation in mathematics to facilitate systematic analysis in political science research. By the end of this course, students will have a basic grasp of mathematical foundations for key analytical methods utilized in political science research. Topics covered will include basic calculus, static optimization theory, and linear algebra.

The course focuses on fundamental mathematics skills required for statistical, quantitative, and formal modeling courses at Stony Brook University. It specifically prepares students for PhD courses in the Methods sequence (Data Analysis I, II, and III), Political Economy, Game Theory, and Formal Models in American Politics. The primary purpose of the course is to refresh some college math basics and introduce key concepts needed for the methods sequence. It is important to note that this math camp is not meant to replace what you would have learned if you had taken college calculus and algebra but rather to highlight and reinforce the critical concepts necessary for success in these advanced courses. So, the math camp can be thought of as a quick and focused review of the *content you should know to read and write math at the level required in the Ph.D. courses*.

Structure

Participation in the math camp is mandatory for all first-year PhD students. While attendance is required, the course will not be graded.

The class is scheduled for the week of August 17th to 21st. Daily meeting blocks differ from day to day because they are built around the mandatory orientations, check-ins, and union meeting described in Section III; please consult the schedule carefully rather than assuming a fixed daily start time. We will cover the material from the book, highlighting key points and expanding on important concepts. In class, we will also work on problems individually to reinforce the material. Students should read the designated chapters on their own and come to class prepared to

address questions and solve problems. This approach aims to expose students to the type of math they can expect in Ph.D. classes and allows for immediate clarification of any questions. The math camp is an ideal time to ask questions, no matter how basic they may seem, ensuring everyone understands how to read and write math effectively. Daily problem sets and exercises will be provided for self-assessment and can be completed as homework, with answers supplied for reference.

It is important to note that learning how to read and write math takes time, especially if a student's exposure to college-level math has been limited. While one week may not be enough to master these skills, it is a valuable opportunity to learn the basics, such as understanding mathematical symbols and reading mathematical statements. If you still feel uncertain after the math camp, you should dedicate regular self-study sessions to the topics covered in the camp that you struggled with and be patient as you develop your skills. The book by **Moore and Siegel is an excellent resource for covering the basics in depth if you need help with reading and writing at a college math level.**¹

II Textbook and Additional Material

Required

There is one required textbook for the course, available online free of charge:

King, Gary et al. 2020. "The Harvard Math Prefresher for Political Scientists." <https://iqss.github.io/prefresher/>

Additional Readings (Optional)

Additional recommended materials include:

[M&S] Moore, Will H, and David A Siegel. 2013. *A Mathematics course for political and social research*. Princeton University Press. YouTube lectures available here: <http://people.duke.edu/~das76/moosiebook.html>

[M&R] Margalit, D., & Rabinoff, J. (2019). *Interactive Linear Algebra*. Georgia Institute of Technology. Retrieved from <https://textbooks.math.gatech.edu/ila/index2.html>

Gill, Jeff. 2006. *Essential mathematics for political and social research*. Cambridge University Press.

Simon, Carl P, and Lawrence Blume. 1994. *Mathematics for economists*. Norton & Company, Inc.

III Schedule

Daily Meeting Blocks

Because the Department and the Graduate School have scheduled several orientations during the same week, the daily blocks are uneven. **No class is held during any mandatory event.** There is a lunch break every day, and each long block is broken by a 15-minute recess. **Please read the table below carefully: the starting time changes from day to day.**

¹If after the Math Camp, you feel you are struggling to read and write math at the required level, I strongly recommend you get the book and use it as reference. Additionally, you could watch the video series based on the same book here: <http://people.duke.edu/~das76/moosiebook.html> (click on "Syllabus with Direct Links to All Videos").

Day	Class Blocks	Institutional Events (no class)
Mon, 17/08	9:00–10:30 a.m. <i>15-min break</i> 10:45 a.m.–12:15 p.m. <i>Lunch 12:15–1:00 p.m.</i> 1:00–3:00 p.m.	GSEU/RAU union meeting, 3:00–3:30 p.m., in our classroom (mandatory)
Tue, 18/08	10:00–11:30 a.m. <i>15-min break</i> 11:45 a.m.–12:45 p.m. <i>Lunch 12:45–1:45 p.m.</i> 1:45–3:45 p.m.	Student check-in with the Graduate Program Coordinator, 9:00–10:00 a.m. (mandatory) A few students may need to step out for about 10 minutes later in the day.
Wed, 19/08	<i>Lunch 11:30 a.m.–12:50 p.m.</i> 12:50–2:50 p.m. <i>15-min break</i> 3:05–4:05 p.m.	Department Orientation, 8:30–10:00 a.m. (mandatory) Graduate School Orientation, 10:00–11:30 a.m. (mandatory)
Thu, 20/08	9:00–10:30 a.m. <i>15-min break</i> 10:45 a.m.–12:15 p.m. <i>Lunch 12:15–1:15 p.m.</i> 1:15–2:15 p.m.	—
Fri, 21/08	9:00–10:30 a.m. <i>15-min break</i> 10:45 a.m.–12:15 p.m. <i>Lunch 12:15–1:15 p.m.</i> 1:15–3:15 p.m.	—

On Monday. The GSEU/RAU union meeting at 3:00 p.m. takes place in our own classroom, so Monday's afternoon block ends at 3:00 p.m.

Topics

Day	Topics	Textbook Chapters	Suggested Readings
Mon, 17/08	Logic, Sets, Basic Operations, Functions, & Limits	Ch. 1-2	[M&S] Ch. 1-4
Tue, 18/08	Calculus I (Derivatives and Extrema)	Ch. 3.1-3.6, 4.1	[M&S] Ch. 5-6, 8
Wed, 19/08	Calculus II (Integrals)	Ch. 3.7-3.10	[M&S] Ch. 7
Thu, 20/08	Linear Algebra	Ch. 6	[M&S] Ch. 12-14
Fri, 21/08	Multivariate Calculus & Optimization	Ch. 4	[M&S] Ch. 15, 16.1-16.2

Note: The allocation of topics to days may shift depending on our progress during the week. Any change to the meeting blocks themselves will be announced in class and by email.

Class Pace

The course recognizes that students enter with a wide range of mathematical backgrounds. Although the schedule is ambitious, new topics will be introduced only once the class has a firm understanding of the material. The primary way comprehension is assessed is through in-class questions and short activities. For that reason, students are strongly encouraged to speak up whenever a concept is unclear—no matter how basic the question may seem. Questions can be asked during lectures for immediate clarification or addressed afterward in office hours or by email.

Note on Readings

Students are expected to read the material prior to each class. The chapters contain challenging mathematical content. The optimal approach to reading math textbooks is to work through derivations with a pencil and paper. Engaging with the derivations and exercises will be extremely beneficial.